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PI Name:	Yodh, Arjun Ph.D.		
Project Title:	Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 3)		
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Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
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	This proposal builds on and continues the research described in Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 2) (Ed. note: see "Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 2)"; grant NNX08AO80G in Task Book). The deep roots of the work derive from an experimental program that was peer-reviewed and accepted for flight on the International Space Station (ISS); the instrumentation at the heart of the original flight proposal for ISS was part of the so-called the Light Microscopy Module (LMM). More recently, a Science Concept Review (SCR) for Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 2) was held in June 2011, and the proposed research was strongly endorsed by the External Review Panel in their report on June 14 2011; thus, the science goals and experimental plans for the present proposal, Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 3), have already been critically reviewed. The current research will be integrated into the hardware developed for the space-flights that are named by variants of the term “ACE” (e.g., ACE-H1). In addition, related complex fluid measurements have been carried out in the Binary Colloid Alloy Test (BCAT) BCAT-series flight		

Task Description:	experiments. Since the original proposal Low Volume Fraction Entropically Driven Colloidal Assembly was submitted and approved, the NASA budget has been in flux and soft matter science has evolved too. In response to these developments, the present investigations are leaner, with smaller budgets, and the science aims have evolved to take advantage of important recent breakthroughs in the field. Herein we describe these complex fluid experiments. Note also, whenever possible, we will be joining forces with our colleagues from Europe to perform collaborative experiments within the ESA and NASA facilities. The overall rationale for these investigations is that the elimination of particle sedimentation effects in microgravity creates a purely “thermodynamic” environment for colloidal suspensions wherein particle size, shape and volume fraction, and interparticle interactions are the dominant determinants of the assembled structures. This feature enables us to make new materials that do not form on Earth and to elucidate the effects of gravity in soft matter science that are masked in ground-based experiments. The Phase 3 version of Low Volume Fraction Entropically Driven Colloidal Assembly explores a range of experimental complex fluids with themes related to and evolved from previous proposal aims. The original studies were centered around entropically driven low volume-fraction self-assembling complex fluids, suspensions composed of an ensemble of large particles, and additional smaller constituents (either a small colloidal particle, a polymer or a micelle). The present studies retain many of the original themes but also has new focus that reflects the evolution of the complex fluid field, as well as new developments in our laboratory. Broadly, the proposed research will elucidate a wide-range of phenomena in condensed matter physics that has implications for phase-diagrams of entropically driven colloidal media, crystallization, and melting mechanisms in three-dimensional (3D) bulk media and in finite volumes, jamming and glass formation in 3D, and, opportunistically, crystallization and melting mechanisms and aging of colloids/soft-matter based on more anisotropic particles and filaments on a variety of length scales. International collaborators are Prof. Luca Cipelletti, Université Montpellier II, Montpellier, France; Prof. Roberto Piazza, Department of Nuclear Engineering, Politecnico di Milano, Milan, Italy; Prof. Willem Kegel, Van 't Hoff Laboratory for Physical and Colloid Chemistry, Debye Research Institute, Utrecht University, Utrecht, The Netherlands.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research can provide fundamental understanding of natural phenomena affected by gravity for increased efficiency and effectiveness of space-based and industrial processes.
	(Ed. note: Compiled from annual report submitted by PI to W. Meyer/NASA GRC--May 2015.) The abstracts that follow summarize the publications and research from January, 2014 through April, 2015. Full citations are listed in the Bibliography section: Still et al., 2014. Particle tracking and displacement covariance matrix techniques are employed to investigate the phonon dispersion relations of two-dimensional colloidal glasses composed of soft, thermoresponsive microgel particles whose temperature-sensitive size permits in situ variation of particle packing fraction. Bulk, B, and shear, G, moduli of the colloidal glasses are extracted from the dispersion relations as a function of packing fraction, and variation of the ratio G/B with packing fraction is found to agree quantitatively with predictions for jammed packings of frictional soft particles. In addition, G and B individually agree with numerical predictions for frictional particles. This remarkable level of agreement enabled us to extract an energy scale for the interparticle interaction from the individual elastic constants and to derive an approximate estimate for the interparticle friction coefficient. Jeong et al., 2014. PNAS. Confined liquid crystals (LC) provide a unique platform for technological applications and for the study of LC properties, such as bulk elasticity, surface anchoring, and topological defects. In this work, lyotropic chromonic liquid crystals (LCLCs) are confined in spherical droplets, and their director configurations are investigated as a function of mesogen concentration using bright-field and polarized optical microscopy. Because of the unusually small twist elastic modulus of the nematic phase of LCLCs, droplets of this phase exhibit a twisted bipolar configuration with remarkably large chiral symmetry breaking. Further, the hexagonal ordering of columns and the resultant strong suppression of twist and splay but not bend deformation in the columnar phase, cause droplets of this phase to adopt a concentric director configuration around a central bend disclination line and, at sufficiently high mesogen concentration, to exhibit surface faceting. Observations of director configurations are consistent with Jones matrix calculations and are understood theoretically to be a result of the giant elastic anisotropy of LCLCs. Basu et al., 2014. We study the rheological behavior of colloidal suspensions composed of soft sub-micron-size hydrogel particles across the liquid–solid transition. The measured stress and strain-rate data, when normalized by thermal stress and time scales, suggest our systems reside in a regime wherein thermal effects are important. In a different vein, critical point scaling predictions for the jamming transition, typical in athermal systems, are tested. The observation suggests that our system exhibits a glass transition near the onset of rigidity, but it also exhibits a jamming-like scaling further from the transition point. These observations are thought-provoking in light of recent theoretical and simulation findings, which show that suspension rheology across the full range of microgel particle experiments can exhibit both thermal and athermal mechanisms. Jeong et al., 2014. Langmuir. We report on the homeotropic alignment of lyotropic chromonic liquid crystals (LCLCs). Homeotropic anchoring of LCLCs is difficult to achieve, and this challenge has limited development of applications for LCLCs. In this work, homeotropic alignment is achieved using noncovalent interactions between the LCLC molecules and various alignment layers including graphene, parylenefilms, poly(methyl methacrylate) films, and fluoropolymer films. The LCLC molecules are unique in that they self-assemble via noncovalent interactions in water into elongated aggregates which, in turn, form nematic and columnar liquid crystal (LC) phases. Here we exploit these same noncovalent interactions to induce homeotropic anchoring of the nematicLCLC. Homeotropic alignment is confirmed by polarized optical microscopy and conoscopy. We also report on novel transient stripe textures that occur when an initial flow-induced planar alignment transforms into the equilibrium homeotropic alignment required by boundary conditions. An understanding of this behavior could be important for switching applications. Lohr et al., 2014. Soft Matter. We explore micropatterned director structures of aqueous lyotropic chromonic liquid crystal (LCLC) films created on square-lattice cylindrical-micropost substrates. The structures are manipulated by modulating the LCLC mesophases and their elastic properties via concentration through drying. Nematic LCLC films exhibit preferred bistable alignment along the diagonals of the micropost lattice. Columnar LCLC films, dried from nematics, form two distinct director and defect configurations: a diagonally aligned director pattern with local squares

Task Progress:	of defects, and an off-diagonal configuration with zig-zag defects. The formation of these states appears to be tied to the relative splay and bend free energy costs of the initial nematic films. The observed nematic and columnar configurations are understood numerically using a Landau-de Gennes free energy model. Among other attributes, the work provide first examples of quasi-2D micropatterning of LC films in the columnar phase and lyotropic LC films in general, and it demonstrates alignment and configuration switching of typically difficult-to-align LCLC films via bulk elastic properties. Yunker et al., 2014. This review collects and describes experiments that employ colloidal suspensions to probe physics in ordered and disordered solids and related complex fluids. The unifying feature of this body of work is its clever usage of poly(N-isopropylacrylamide) (PNIPAM) microgel particles. These temperature-sensitive colloidal particles provide experimenters with a 'knob' for in situ control of particle size, particle interaction and particle packing fraction that, in turn, influence the structural and dynamical behavior of the complex fluids and solids. A brief summary of PNIPAM particle synthesis and properties is given, followed by a synopsis of current activity in the field. The latter discussion describes a variety of soft matter investigations including those that explore formation and melting of crystals and clusters, and those that probe structure, rearrangement and rheology of disordered (jammed/glassy) and partially ordered matter. The review, therefore, provides a snapshot of a broad range of physics phenomenology which benefits from the unique properties of responsive microgel particles. Lohr et al., 2014. Physical Review E. We investigate the vibrational modes of quasi-two-dimensional disordered colloidal packings of hard colloidal spheres with short-range attractions as a function of packing fraction. Certain properties of the vibrational density of states (vDOS) are shown to correlate with the density and structure of the samples (i.e., in sparsely versus densely packed samples). Specifically, a crossover from dense glassy to sparse gel-like states is suggested by an excess of phonon modes at low frequency and by a variation in the slope of the vDOS with frequency at low frequency. This change in phonon mode distribution is demonstrated to arise largely from localized vibrations that involve individual and/or small clusters of particles with few local bonds. Conventional order parameters and void statistics did not exhibit obvious gel-glass signatures as a function of volume fraction. These mode behaviors and accompanying structural insights offer a potentially new set of indicators for identification of glass-gel transitions and for assignment of gel-like versus glass-like character to a disordered solid material. Dreyfus et al., 2015. Hyperuniformity characterizes a state of matter for which (scaled) density fluctuations diminish towards zero at the largest length scales. However, the task of determining whether or not an image of an experimental system is hyperuniform is experimentally challenging due to finite-resolution, noise, and sample-size effects that influence characterization measurements. Here we explore these issues, employing video optical microscopy to study hyperuniformity phenomena in disordered two-dimensional jammed packings of soft spheres. Using a combination of experiment and simulation we characterize the possible adverse effects of particle polydispersity, image noise, and finite-size effects on the assignment of hyperuniformity, and we develop a methodology that permits improved diagnosis of hyperuniformity from real-space measurements. The key to this improvement is a simple packing reconstruction algorithm that incorporates particle polydispersity to minimize the free volume. In addition, simulations show that hyperuniformity in finite-sized samples can be ascertained more accurately in direct space than in reciprocal space. Finally, our experimental colloidal packings of soft polymeric spheres are shown to be effectively hyperuniform. Jeong et al., 2015. We study chiral symmetry-broken configurations of nematic liquid crystals (LCs) confined to cylindrical capillaries with homeotropic anchoring on the cylinder walls (i.e., perpendicular surface alignment). Interestingly, achiral nematic LCs with comparatively small twist elastic moduli relieve bend and splay deformations by introducing twist deformations. In the resulting twisted and escaped radial (TER) configuration, LC directors are parallel to the cylindrical axis near the center, but to attain radial orientation near the capillary wall, they escape along the radius through bend and twist distortions. Chiral symmetry-breaking experiments in polymer-coated capillaries are carried out using Sunset Yellow FCF, a lyotropic chromonic LC with a small twist elastic constant. Its director configurations are investigated by polarized optical microscopy and explained theoretically with numerical calculations. A rich phenomenology of defects also arises from the degenerate bend/twist deformations of the TER configuration, including a nonsingular domain wall separating domains of opposite twist handedness but the same escape direction and singular point defects (hedgehogs) separating domains of opposite escape direction. We show the energetic preference for singular defects separating domains of opposite twist handedness compared with those of the same handedness, and we report remarkable chiral configurations with a double helix of disclination lines along the cylindrical axis. These findings show archetypally how simple boundary conditions and elastic anisotropy of confined materials lead to multiple symmetry breaking and how these broken symmetries combine to create a variety of defects.

Bibliography Type:	Description: (Last Updated: 06/16/2025)
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